# Year 13 Maths - Numerical integration

Similar questions - Trapezium rule and bounds

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1**

**(a)** Use the trapezium rule with 4 strips of equal width to estimate $\int_0^2 e^{-x^2}\,dx$, giving your answer to 3 significant figures. **[4]**

**(b)** The graph of $y = e^{-x^2}$ is convex (curves upwards) for $x > \frac{1}{\sqrt{2}}$ and concave for $0 < x < \frac{1}{\sqrt{2}}$. Explain why it is not clear from the graph whether your answer to (a) is an over- or under-estimate. **[2]**

**(c)** State how the estimate could be made more accurate. **[1]**

**2** Priya investigates the area $A$ under $y = \frac{1}{x}$ between $x = 1$ and $x = 2$ using rectangles of width 0.25.

**(a)** Show that the lower and upper bounds for $A$ obtained from four rectangles are 0.635 and 0.760 respectively, to 3 decimal places. **[4]**

**(b)** Explain how Priya could obtain a closer estimate using rectangles. **[1]**

**(c)** Write down, in terms of $y$ and $\delta x$, an expression for the exact value of $A$, and state its exact value. **[3]**

**Question 1**

(a) $h = 0.5$: $y(0) = 1$, $y(0.5) = 0.7788$, $y(1) = 0.3679$, $y(1.5) = 0.1054$, $y(2) = 0.0183$. $\frac{0.5}{2}[1 + 0.0183 + 2(0.7788 + 0.3679 + 0.1054)]$ = 0.881 (3 s.f.)

(b) The curve changes from concave to convex inside the interval, so trapezia lie below the curve on the concave part and above it on the convex part; the two errors partly cancel in an unknown way

(c) Use more (narrower) strips

**Question 2**

(a) Upper bound uses left-hand heights $1, \frac{4}{5}, \frac{2}{3}, \frac{4}{7}$: $0.25(1 + 0.8 + 0.6667 + 0.5714) = 0.760$. Lower bound uses right-hand heights $\frac{4}{5}, \frac{2}{3}, \frac{4}{7}, \frac{1}{2}$: $0.25(0.8 + 0.6667 + 0.5714 + 0.5) = 0.635$

(b) Use narrower rectangles (smaller $\delta x$); the bounds then get closer together

(c) $A = \lim_{\delta x \to 0} \sum y\,\delta x$ (summed from $x = 1$ to $x = 2$) $= \displaystyle\int_1^2 \frac{1}{x}\mathrm{d}x = \ln 2$